

RFM Customer Segmentation Dashboard Documentation

1. Dashboard Objective

The objective of this dashboard is to visualize the results of RFM (Recency, Frequency, Monetary) analysis and present customer segmentation, revenue contribution, and customer behavior in a clear and business-ready format. The dashboard helps identify high-value customers, revenue-driving segments, and potential churn risks.

2. Tools & Technologies Used

- Power BI Desktop – for dashboard development and visualization
- Microsoft Excel – for cleaned input dataset
- DAX – for measures, calculated columns, and Top-N logic

3. Data Description

The dashboard is built using a cleaned customer-level dataset containing purchase behavior information. Key fields include Customer ID, Recency, Frequency, Monetary value, RFM scores, and customer segment labels. The dataset consists of 329 unique customers with a total revenue of approximately **103.9K**.

4. RFM Methodology

Customers were evaluated using the RFM framework: Recency measures how recently a customer purchased, Frequency measures how often purchases occur, and Monetary measures total spend. Quantile-based scoring was applied to each metric, and combined RFM scores were mapped to business-friendly segments such as Champions, Loyal Customers, New Customers, At Risk, Lost Customers, and Others.

5. Dashboard Structure

5.1 KPI Section: Displays Total Customers, Total Revenue, and Average Monetary Value to provide a quick business overview.

5.2 Segmentation & Revenue Analysis: Includes customer segment distribution and revenue by segment visuals to identify dominant and high-value customer groups.

5.3 Behavioral & Customer-Level Analysis: Contains a Frequency vs Monetary scatter plot and a Top 10 Customers table to analyze purchasing behavior and identify high-value individuals.

6. Measures Used (DAX)

Key measures created using DAX include:

- Total Customers – Distinct count of Customer ID
- Total Revenue – Sum of Monetary value
- Average Monetary – Average spend per customer
- Total Monetary – Measure used for Top-N customer ranking

7. Visual Functionality

- KPI Cards provide high-level performance indicators.
- Bar and Column charts show customer distribution and revenue contribution by segment.
- Scatter plot visualizes customer behavior using average frequency and monetary values, with bubble size representing customer count.
- Top 10 Customers table identifies the highest revenue-generating customers using measure-based Top-N filtering.

8. Key Insights

Champions and Loyal Customers contribute the highest revenue and demonstrate strong engagement. Loyal Customers form the largest customer base, while At Risk and Lost Customers show low engagement and spending. A small number of customers generate a disproportionate share of revenue, highlighting the importance of retention strategies.

9. Business Recommendations

The business should prioritize retention and loyalty programs for high-value segments, run re-engagement campaigns for At Risk customers, and nurture New Customers to increase purchase frequency and long-term value.

10. Conclusion

This RFM Customer Segmentation Dashboard effectively translates analytical results into actionable business insights. It demonstrates the ability to perform customer segmentation, apply DAX logic, and design a structured, professional Power BI dashboard suitable for business decision-making.